Grade 7	Name____________________
Unit 5 Assessment	Date_________________
Calculator	Period__________

Showing Work with a Calculator: Because you are using a calculator, it is possible to accidently punch the wrong button. To ensure you earn partial credit, show the operation(s) that you entered into the calculator and the result. It is helpful if you label the operation(s) with information from the word problem.

1. An architect builds a scale model of a building he is designing. The scale model is $\frac{1}{40}$ of the original size of the building. The height, in inches (in.), of the scale model is 16 in.

   Determine the height, in inches, of the building. Show your work.

work	
Height of Building (in.)	640

2. Talking or texting on a cell phone can be a distraction. A survey of 80 middle schoolers at Sebastian Middle School found that 60% of those surveyed had someone who was talking or texting walk into them. Another survey conducted by a cell phone company found that of 738 of cell phone owners, 23% of those surveyed had someone who was talking or texting walk into them.

a. What are the two populations that were surveyed?

Populations	Middle schoolers at Sebastian Middle School and cell phone owners

b. Complete the statement.

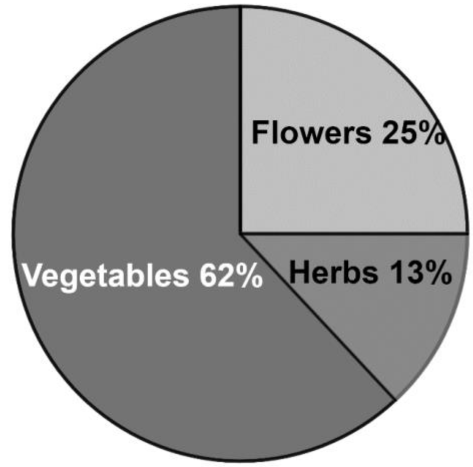
- ○ Both surveys
- ○ Neither survey
- ○ The survey conducted by a cell phone company
- ○ The survey conducted at Sebastian Middle School

would be a representative sample of all cell phone users.

3. After a new process of producing a computer chip was introduced at a manufacturing plant, a random sample of 640 of the computer chips showed that 8 of the chips had a defect. Determine the number of defective computer chips when $100{,}000$ computer chips are manufactured.

work	
Defective Computer Chips	1250

4. In 2015, a seed company surveyed people to ask them what type of seeds they plan to buy. The results are shown.



a. The population of Yulee, Florida in 2015 was $11{,}672$. Use the results to predict the number of people in Yulee who may have bought flower seeds in 2015.

work	
People in Yulee Who May Have Bought Flower Seeds	2918

b. The population of Eustis, Florida was $19{,}200$ in 2015. Use the results to predict the number of people in Eustis who may have bought vegetable seeds in 2015.

work	
People in Eustis Who May Have Bought Vegetable Seeds	11904

5. A deck of cards consists of 80 cards divided into 8 different sets of colors and each color has cards numbered 1 to 10. The colors are red, blue, green, yellow, orange, purple, pink, and white. The table shows the results from a simulation of drawing a random card from the deck, where each tally mark represents one occurrence.

<table>
<tr><th></th><th colspan="10">Card Number</th><th rowspan="2">Total</th></tr>
<tr><th>Color</th><th>1</th><th>2</th><th>3</th><th>4</th><th>5</th><th>6</th><th>7</th><th>8</th><th>9</th><th>10</th></tr>
<tr><td>Red</td><td>|||</td><td>|</td><td></td><td>||</td><td>|</td><td>||</td><td></td><td>|</td><td>|</td><td></td><td>11</td></tr>
<tr><td>Blue</td><td>||</td><td>|</td><td>||</td><td>|</td><td>|</td><td>|</td><td>||</td><td></td><td>|</td><td>||</td><td>13</td></tr>
<tr><td>Green</td><td></td><td>||</td><td>|</td><td>|</td><td>||</td><td>|</td><td>|</td><td>||</td><td></td><td>|</td><td>11</td></tr>
<tr><td>Yellow</td><td>|</td><td>|</td><td>||</td><td>||</td><td>|</td><td>||</td><td>|</td><td>|</td><td>||</td><td>||</td><td>15</td></tr>
<tr><td>Orange</td><td>|</td><td></td><td>|</td><td>|</td><td>|</td><td></td><td></td><td>|</td><td>|</td><td></td><td>6</td></tr>
<tr><td>Purple</td><td></td><td>|</td><td></td><td>|</td><td>||</td><td>|</td><td>|</td><td></td><td></td><td>|</td><td>7</td></tr>
<tr><td>Pink</td><td>|</td><td>|</td><td>|</td><td>|</td><td>|</td><td>|</td><td>||</td><td>|</td><td>||</td><td>|</td><td>12</td></tr>
<tr><td>White</td><td>|</td><td>||</td><td>|</td><td></td><td>||</td><td>||</td><td>|</td><td>|||</td><td>||</td><td>|</td><td>15</td></tr>
<tr><td>Total</td><td>9</td><td>9</td><td>8</td><td>9</td><td>11</td><td>10</td><td>8</td><td>9</td><td>9</td><td>8</td><td>90</td></tr>
</table>

a. Based on the simulation what is the probability of drawing a yellow card? Write your answer using a fraction.

Yellow Card	$\frac{15}{90}$ or $\frac{3}{18}$ or $\frac{1}{6}$

b. Based on the simulation what is the probability of drawing a card with the number 4? Write your answer using a fraction.

Card with 4	$\frac{9}{90}$ or $\frac{3}{30}$ or $\frac{1}{10}$

c. Use the results from the simulation to predict the number of times a pink card would be drawn when the total number of cards drawn in the simulation is 585.

Number of Pink Cards	78

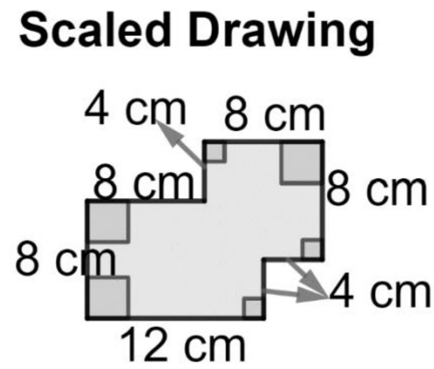
d. Use the results from the simulation to predict the number of times a card with the number 7 would be drawn when the total number of cards drawn in the simulation is $1{,}440$.

Number of Cards with 7	128

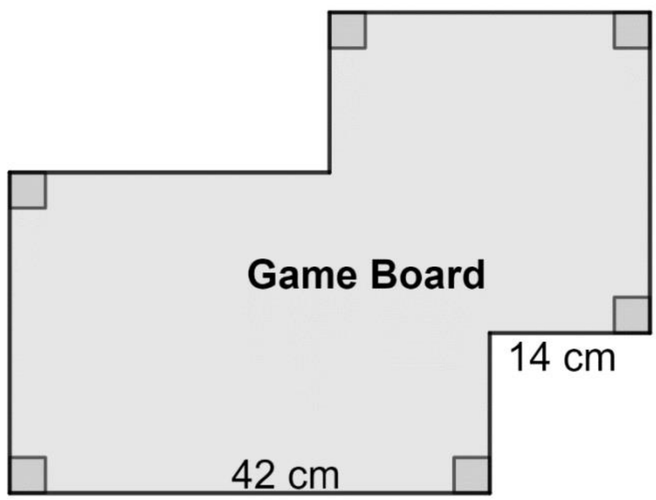
6. Each participant in the Santos Fat Tire Festival in Ocala rides 50 miles of mountain bike trails. The map uses a scale where 1 inch represents 0.25 miles.
Determine the length, in inches (in.), the trails are on the map.

work	
Length of Mountain Bike Trails on the map (in.)	200

7. The scaled drawing of a game board, where measurements are given in centimeters (cm) is shown.



A drawing of the game board is shown where some of the measurements are shown.



a. Determine the scale factor that can be used to determine the perimeter of the game board.

Scale Factor for the Perimeter	3.5 or $\frac{7}{2}$

b. Determine the scale factor that can be used to determine the area of the game board.

Scale Factor for the Area	3.5^2 or 12.25 or $\left(\frac{7}{2}\right)^2$ or $\frac{49}{4}$